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FURTHER PALEOMAGNETIC RESULTS FOR THE SAN JUAN VOLCANIC FIELD OF SOUTHERN COLORADO

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Combining paleomagnetic data for 17 new sites from the northwest portion of the (Oligocene) San Juan volcanic field of southern Colorado with data for 29 sites previously published yields a paleomagnetic pole at 85°N, 114°E (with a 95% confidence circle of 7.5° radius). A further combination of the San Juan data with the results of other studies on rocks of Oligocene age from tectonically stable parts of North America gives a mid-Tertiary reference pole located at 81°N, 132.5°E, with a confidence circle of approximately 4°. Mid-Tertiary paleomagnetic poles for the western edge of the continent diverge markedly from this reference pole.

1. Introduction

The San Juan volcanic field is a well-exposed, largely unaltered, geologically complex assemblage of lavas, welded tuffs, volcanoclastic sedimentary rocks, and hypabyssal intrusive rocks that covers roughly 25,000 km^2 in the Rocky Mountains of southern Colorado and northern New Mexico (Fig. 1). Recent fields studies and K-Ar age determinations by personnel of the U.S. Geological Survey make it a tempting target for the paleomagnetist. The history of the volcanic field, summarized by Steven and Lipman [1] and by Lipman et al. [2], begins in the early Oligocene with intermediate volcanism from scattered volcanic centers, succeeded about 30 m.y. ago by ash-flow eruptions that accompanied doming, collapse, and resurgence of at least 15 major calderas. Welded tuff units from this phase of volcanic activity extend widely over the field; at least 17 separate ash-flow sheets are known, yielding K-Ar dates ranging roughly from 30 to 22.5 m.y. B.P. Following the caldera phase of igneous activity, eruption of a bimodal suite of basalt and rhyolite continued, at a much reduced rate, until the Pliocene. These successive volcanic stages are unmistakable petrologically, although they overlap one another extensively in both time and space.

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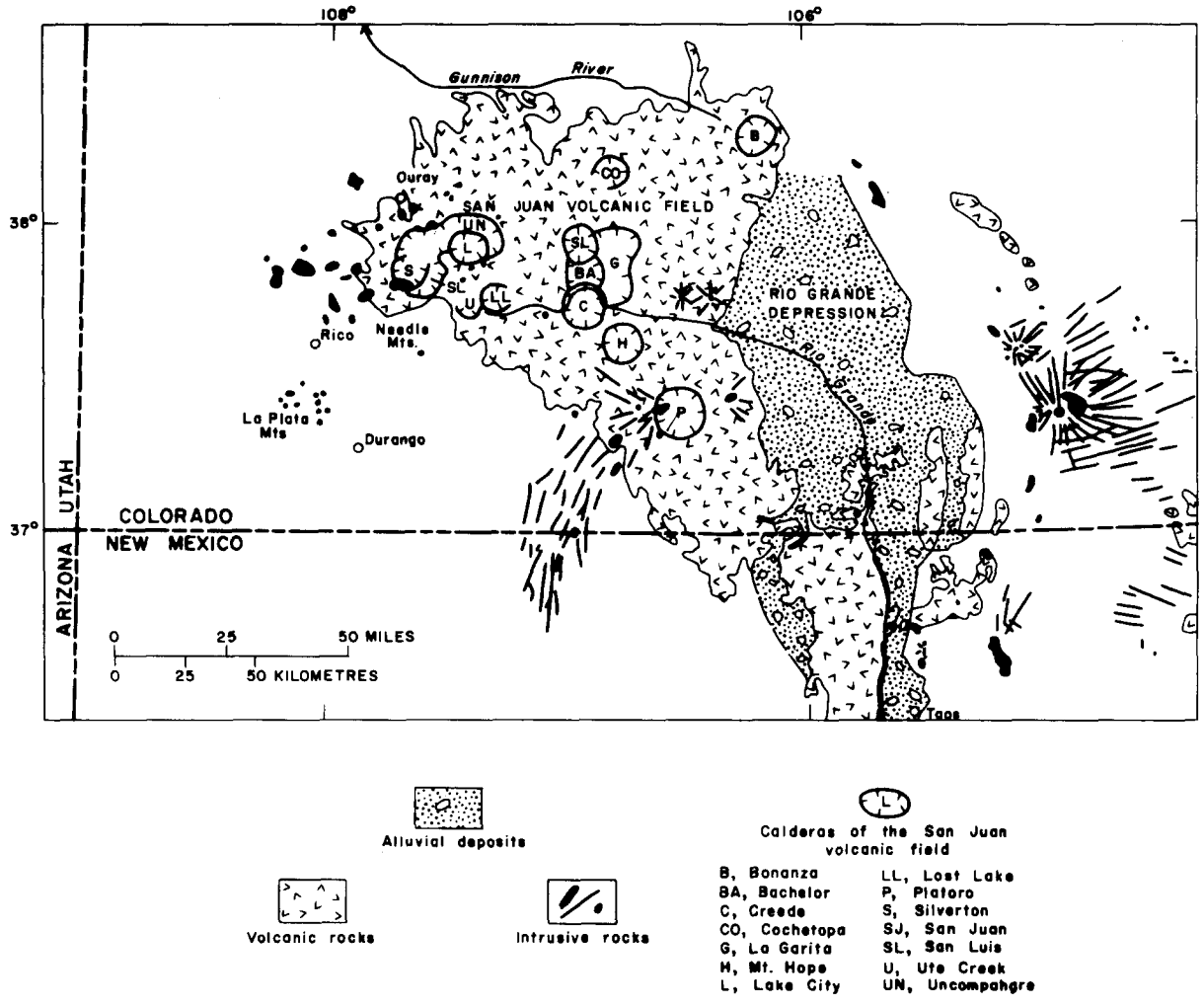


Fig. 1. Index map, modified from Lipman et al. [2].

The first paleomagnetic field work in the San Juan volcanic field was by Tanaka and Kono [3,4], with field guidance from one of us (P.W.L.). They made a reconnaissance survey of 16 sites scattered over much of the field. Subsequently, Diehl et al. [5] have completed a more detailed survey of the southeastern portion of the field, and S.D. Sheriff (M.S. thesis, Western Washington State college, 1976) has similarly investigated the younger calderas to the northwest. The purpose of this communication is to summarize the paleomagnetic work, which is largely complete. A subsequent paper will deal with aspects of paleosecular variation, the reversal sequence, and magnetic intensities. Not included in the present summary is work by Peter Graebner (M.S. thesis, Colorado School of Mines, 1968) on natural remanent magnetism of the Thirtynine Mile volcanic field, located about 80 km northeast of the San Juan exposures. Rocks of the Thirtynine Mile field are similar petrologically to those in the San Juan field and are of approximately the same age. Graebner's work establishes the magnetic stability of the Thirtynine Mile field and defines several polarity transitions, but no mean directions or paleomagnetic pole positions were calculated because the rocks were not magnetically cleaned. Other relevant paleomagnetic studies from the same general

region include Larsen and Strangway's [6] work on the Spanish Peaks dike swarm (late Oligocene/early Miocene) of Colorado and New Mexico, the summary by Grommé and McKee [7] of paleomagnetic data for mid-Tertiary (22–37 m.y.) volcanic rocks from Utah, Nevada, Oregon, and California, and an investigation of Pliocene volcanics from the Rio Grande Gorge, New Mexico, by Ozima et al. [8].

2. Summary of paleomagnetic data

Table 1 summarizes paleomagnetic data by Tanaka and Kono [3] (TK), Diehl et al. [5] (DO), and Sheriff (SS). The TK entry has been recalculated to eliminate the post-Oligocene (22.5 m.y.) Sunshine Creek Tuff (site SJ 14), the Fish Canyon Tuff (site SJ 7) for which Tanaka and Kono list an anomalously low inclination (2.9°), and the Sapinero Mesa Tuff (site SJ 15) which, as will be explained below, has an anomalous direction of remanent magnetization and may be misidentified. The DO data also have been recalculated to incorporate slightly revised tectonic (tilt) corrections for the Fish Canyon Tuff and Nelson Mountain Tuff entries. In addition, sites SS 34, 35, and 37 (and Tanaka and Kono's site 15) are not included in these calculations because these four sites, variously identified as Sapinero Mesa Tuff and Dillon Mesa Tuff, define a tight group some 50° from the mean of other sites and may represent an unrecognized ash-flow unit magnetized during a polarity transition, an "excursion", or other period of abnormal behavior of the geomagnetic field. The Sheriff data, because never previously reported, are presented in detail in Table 2. Sampling and laboratory methods employed by Sheriff were substantially those described in the DO paper [5], except for use of the sun compass for orientation.

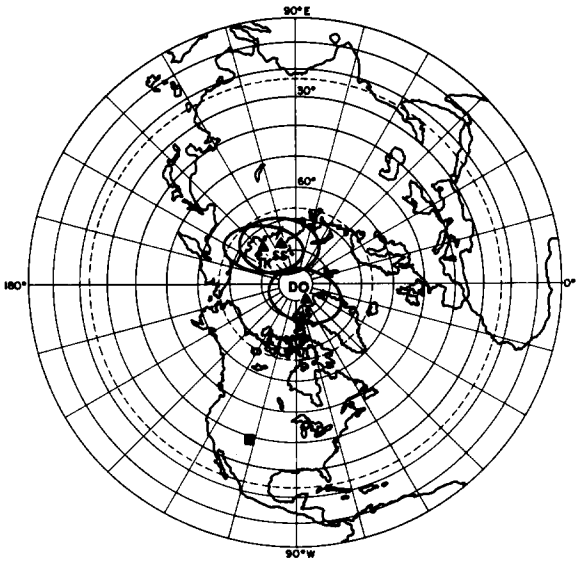


Fig. 2. Paleomagnetic poles for the San Juan volcanics. TK = Tanaka and Kono [3]; DO = Diehl et al. [5]; SS = S.D. Sheriff (M.S. thesis, Western Washington State College, 1976). Ovals are 95% confidence limits. Square indicates sampling location.

As shown in Fig. 2, there is excellent agreement between the SS and TK poles, but the DO data agree less well with the other two studies.

TABLE 1

Paleomagnetic directions and pole locations for the rocks of the San Juan volcanic field of southern Colorado

Author	N	Dec. (°)	Inc. (°)	α_{95} (°)	k	Lat. °N	Long °E	δp (°)	δm (°)
TK	11	165.6	−47	9.9	22.34	74.5	129	8.3	12.8
DO	18	185	−59.5	7.2	24.08	85	308	8.1	10.8
SS	17	171	−45	10.6	12.27	76	109	8.5	13.4

TK = recalculated from Tanaka and Kono [3]; DO = recalculated from Diehl et al. [5]; SS = recalculated from S.D. Sheriff (M.S. thesis, Western Washington State College, 1976). "Dec." and "Inc." refer to mean directions of remanent magnetism after AF demagnetization; Lat. and Long. refer to paleomagnetic pole locations. N = number of sites. α_{95} = radius of circle of confidence at the 95% confidence level [9]; k = precision parameter [9]; δp, δm = semi-axes of oval of confidence about mean pole. (All entries are of mixed polarity, although reverse polarities dominate.)

TABLE 2

Remanent magnetism of Oligocene rocks from the northwestern part of the San Juan volcanic field, Colorado (after S.D. Sheriff, M.S. thesis, Western Washington State College, 1976)

Site	Formation	N	Dec. (°)	Inc. (°)	α_{95} (°)	k	Lat. °N	Long. °W	Age (m.y.)
51	Fisher Quartz Latite	6	345.5	53.0	3.2	430.4	37.74	106.83	26.4
20	Lavas of Uncompaghre Peak	6	197.5	−27.5	5.3	163.5	38.03	107.27	
19		4	201.0	−25.5	4.3	466.2	38.03	107.27	
18		4	185.5	−45.0	4.5	417.2	38.01	107.29	
41	Crystal Lake Tuff	9	10.0	40.0	8.8	35.1	37.79	107.14	
8	Fish Canyon Tuff	7	20.5	48	2.7	487.8	37.99	107.29	27.8
40		7			4.9	150.6	37.79	107.18	
42		6			2.0	1084.1	37.79	107.16	
46	Pyroxene andesite	6	168.5	−27.0	5.6	142.0	37.98	107.57	
1	Quartz monzonite intrusion (post-Uncompaghre Peak)	5	346.0	56.0	4.8	254.9	37.92	107.33	
2		6			2.6	678.4	37.92	107.33	
3		6			5.2	165.9	37.92	107.34	
4	Burns Member of Silverton volcanics	10	185.0	−56.0	2.7	328.0	38.08	107.34	
15									
36	Sapinero Mesa Tuff	5	168.0	−21.0	3.9	390.1	38.03	107.15	
6	Eureka Member of Sapinero Mesa Tuff, ponded in caldera	6	187.0	−49.0	5.6	142.4	38.01	107.31	
16		6			3.8	317.7	38.03	107.32	
5		4	144.5	−56.0	5.2	312.6	38.02	107.36	
17		7			3.1	374.9	38.02	107.33	
64	Dillon Mesa Tuff	6	125.0	−76.0	2.9	526.4	38.41	107.32	
65	Sapinero Mesa Tuff	7			2.7	500.5	38.41	107.32	
66	Blue Mesa Tuff	7	147.5	−66.0	6.5	86.9	38.42	107.33	
14	Ute Ridge Tuff	7	145.0	−34.5	5.8	111.3	38.22	107.38	28.4
57	Pre-caldera andesite	7	144.5	−14.5	4.1	214.9	38.13	107.29	
54		6	146.0	−31.0	5.9	128.2	38.09	107.29	

Lat., Long. are location of site. N is number of samples. Other symbols explained in Table 1.

3. Interpretation

The lack of agreement on the part of the DO pole position is puzzling. The data for all three pole locations are based on directions of remanent magnetization averaged over approximately 10^7 years, so there is scant possibility that transient elements of the geomagnetic field have not been eliminated. The DO data are based on magnetic orientations, whereas the TK and SS studies used a sun compass for orientation, so there exists the possibility that local magnetic anomalies have affected the DO orientations to some extent. However, during the sampling stage of the DO investigation considerable care was taken to check the magnetic declinations by taking back azimuths with two magnetic compasses placed ten or more meters apart. Declination anomalies of more than a few degrees therefore are unlikely, and of course the steeper mean

TABLE 3

Mean direction of remanent magnetization and paleomagnetic pole position for Oligocene rocks from the San Juan volcanic field, Colorado. Mean pole position is the vector mean of virtual geomagnetic pole position for each site

N	Mean direction				Mean pole			
	P	I	α_{95}	k	Lat.	Long.	A_{95}	K
36	175.5°	−51°	6.5°	14.44	85°N	114°E	7.5°	11.32

inclination of the DO study cannot be accounted for at all in this way. Another possible (but very unlikely) cause of the partial lack of agreement between the results of the DO study and the other two studies is uncertainty in the tectonic correction. For instance, if a persistent but unrecognized dip of 15–20° to the southeast were to exist in the DO field area (or a dip of similar magnitude to the northwest elsewhere in the field), the discrepancy might thereby be resolved. Some units, particularly thick felsic flows and intrusive rocks, have uncertainties in their attitudes that may range as high as 10°, but in general the attitudes of the thinner flows, ash-flow tuff units, and interbedded sedimentary rocks are reasonably well determined. A structural explanation therefore can almost certainly be ruled out. The most likely explanation for the poor agreement in location between the DO study and the location of the other two poles for the San Juan volcanic field lies in the fact that the rocks in the DO field area, predominantly the southeastern part of the field, tend to be older than units exposed elsewhere. Part of the discrepancy then could be related to real motion of the paleomagnetic pole relative to the sampling area during the Oligocene. This possibility will be investigated in a subsequent paper. At any rate, there exists no obvious reason why the data from the three studies cannot be combined (Table 3) to give a paleomagnetic pole position for the Oligocene of southern Colorado.

In constructing Table 3 multiple results for thin, presumably quickly cooled volcanic units such as ash-flow sheets and single lava flows have been combined, so that only thick felsic flows, caldera fillings, and intrusive units are represented by more than a single entry. Of the 36 sites included in the calculation, 27 are of reverse polarity. As shown by Fig. 3, the San Juan Oligocene pole position is not significantly different from the present rotation axis at the 95% probability level. Since, as shown by Wilson [10] and discussed by Cox [11], there is a persistent tendency throughout at least the later Cenozoic for paleomagnetic poles to lie a few degrees beyond the true location of the geographic pole, on the side opposite the sampling site, the true San Juan pole may lie even closer to the present pole than shown in Fig. 3

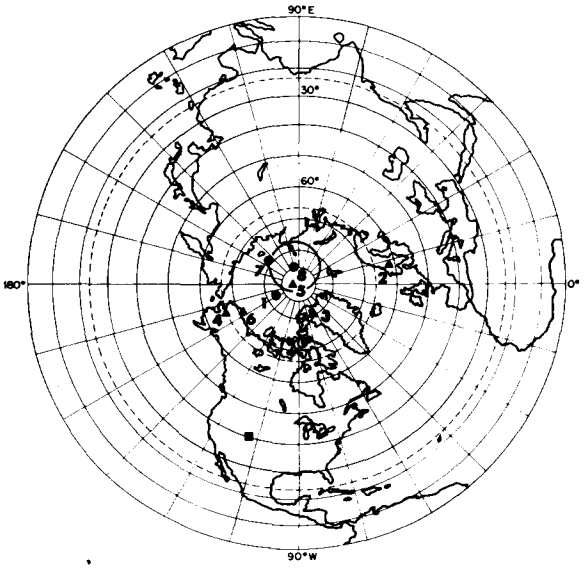


Fig. 3. Middle Tertiary paleomagnetic poles for North America. 1 = Spanish Peaks dikes [6]; 2 = Marys Peak sill [13]; 3 = Western Cascade series [14]; 4 = Mt. Washington Complex [15]; 5 = Hope plutonic complex [16]; 6 = Sooke Gabbro [17]; 7 = Middle Tertiary volcanic units from Utah, Nevada, Oregon and California [7]; 8 = San Juan volcanics (this paper).

4. Mean Oligocene pole

Also in Fig. 3 the San Juan pole position is compared with other mid-Tertiary pole positions for North America. Included here are only those studies that embraced a substantial quantity of Oligocene rocks; pole locations for solely upper Eocene or lower Miocene units, for example, are not included. In Fig. 3, poles from the western portion of the North American Cordillera are shown by triangles, while pole positions from further east (three only) are shown by circles. The three positions from outside the westernmost Cordillera, together with pole 5 from south-central British Columbia, form a reasonably tight group near the present geographic pole, whereas the remaining four poles are scattered widely. Particularly striking is the non-random, almost linear distribution of Oligocene pole positions from the western Cordillera. As one of us [12] has shown recently, paleomagnetic pole position from the western edge of the continent tend to be poorly grouped and so distributed as to suggest systematic large-scale tectonic disturbances, perhaps complicated by irregularities in the geomagnetic field ("excursions") or other sources of scatter. At any rate, it seems at present that the most conservative course in computing a mean Oligocene pole for North America is to combine poles 1, 7 and 8, giving unit weight to each site. The mean pole position so calculated lies at 81°N and 132.5°E. Confidence limits for this pole position are difficult to establish, but again it would appear that the most conservative course is to compute the 95% confidence circle using for dispersion the angular standard deviation of San Juan virtual geomagnetic poles (23.9°), since it is reasonably certain that each San Juan virtual pole position is an independent measurement of the Oliogocene geomagnetic field. Grommé and McKee [7] reported almost precisely this same figure. Using this value of dispersion and assuming a Fisherian distribution of pole locations and roughly 100 independent measurements of the magnetic field, the radius of the 95% confidence circle is approximately 4°. It should be noted that this mean Oligocene pole is based on rock units from a relatively small portion of the southern Cordillera, although there is no obvious reason to believe that it is not equally valid for other parts of North America exclusive of the extreme western edge of the continent.

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